

1 Solution to Problem1

1.1 Dataset Description

The dataset used in this homework is the **MovieLens 100K** dataset, a widely-used benchmark dataset in the field of recommender systems. It consists of 100,000 ratings from 943 users on 1,682 movies. The ratings are on a discrete integer scale from 1 to 5, where 1 indicates strong dislike and 5 indicates strong preference.

The core data file is `u.data`. Each row in the file contains the following four fields:

- `user_id`: an integer between 1 and 943, identifying the user
- `item_id`: an integer between 1 and 1682, identifying the movie
- `rating`: an integer between 1 and 5, representing the user's rating
- `timestamp`: a UNIX timestamp indicating when the rating was made

1.2 Data Preprocessing

To better understand the structure and distribution of the MovieLens 100k dataset before collaborative filtering, we conducted several visualizations. These visualizations help identify patterns, biases, sparsity, and user behaviors in the data.

We plotted a histogram of all ratings using a bar chart, where the x-axis represents rating values from 1 to 5, and the y-axis shows the frequency of each rating.

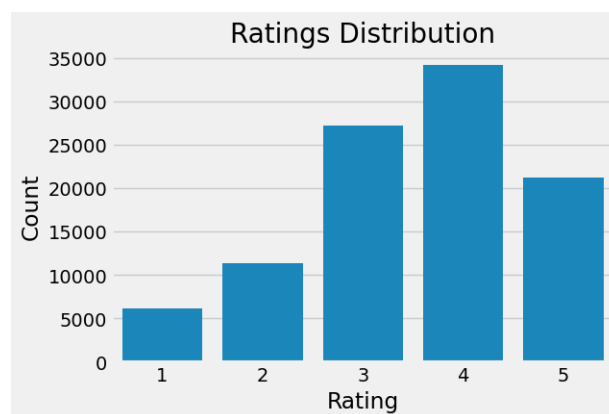


Figure 1: Histogram of Ratings

The distribution is skewed towards medium and higher ratings, with 3 and 4 being the most common. This indicates that users tend to rate movies positively. The low number of ratings below 3 suggests less negative feedback.

We grouped the ratings by user ID and counted the number of ratings each user provided. A histogram was plotted to show the distribution of these counts.

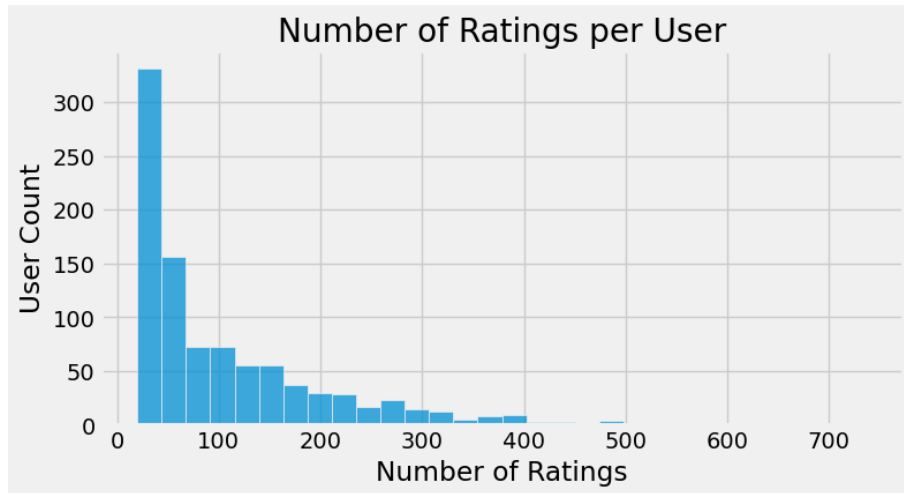


Figure 2: Histogram of Number of Ratings per User

There is a long-tail distribution where most users have rated fewer than 150 movies, but a small number of users are highly active. This uneven distribution implies that some users have much more influence in collaborative filtering models, which should be accounted for in evaluation.

We grouped the ratings by item (movie) ID and counted how many ratings each movie received. A histogram was plotted to visualize the distribution of rating counts across movies.

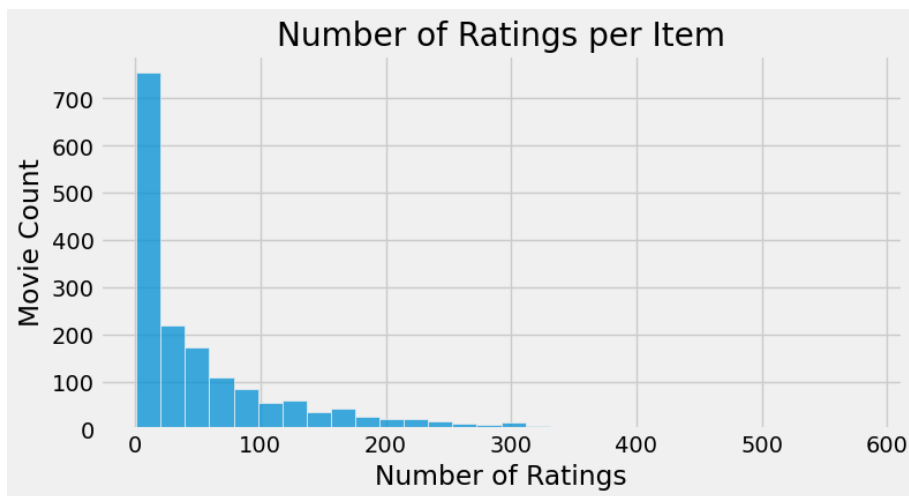


Figure 3: Histogram of Number of Ratings per Movie

Similar to users, the number of ratings per movie also follows a long-tail distribution. A few movies are extremely popular (i.e., frequently rated), while many receive very few ratings. This sparsity among less popular items can negatively impact recommendation quality for cold-start items.

1.3 User-Based Collaborative Filtering Prediction

We performed user-based collaborative filtering to estimate user 1's rating for item 508 using the following steps:

1. **Utility Matrix Construction:** A user-item rating matrix was created by pivoting the dataset, where rows represent users, columns represent items, and cells contain ratings.
2. **Centering the Ratings:** For each user, their average rating was subtracted from all rated items to normalize individual rating scales and account for rating bias.
3. **Filling Missing Values:** Missing ratings (NaNs) were replaced with zeros in the centered matrix to enable cosine similarity computation without introducing bias from non-overlapping items.
4. **Cosine Similarity Calculation:** Cosine similarity was computed between all user vectors based on the normalized (centered and zero-filled) ratings matrix.
5. **Top 10 Similar Users:** The 10 users most similar to user 1 were identified by selecting the highest cosine similarity values, excluding user 1 themselves.
6. **Rating Aggregation for Prediction:** The ratings given by these similar users for item 508 were extracted. The expected rating for user 1 was calculated as the mean of these ratings, ignoring missing values.

Figure 4 shows the cosine similarity scores of the 10 most similar users to user 1.

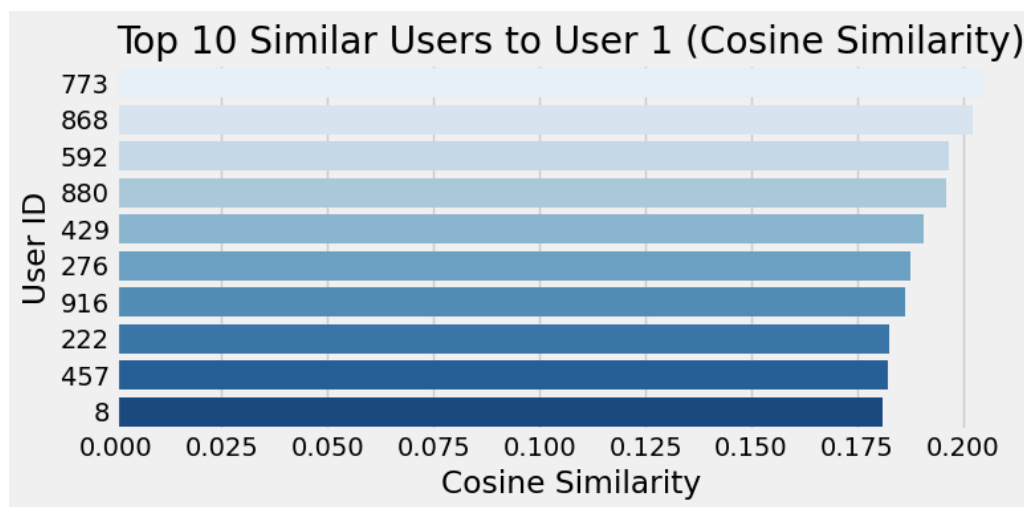


Figure 4: Top 10 Similar Users to User 1

The similarity values range from approximately 0.18 to 0.2, indicating moderate alignment in rating behavior. This suggests that while these users do not have extremely high similarity to user 1, they are the closest available in the dataset based on shared rating patterns. These top similar users form the basis for predicting user 1's rating for other items, such as item 508.

We constructed the user-item rating matrix and visualized the presence of ratings using a binary plot where each dot represents a non-missing rating.

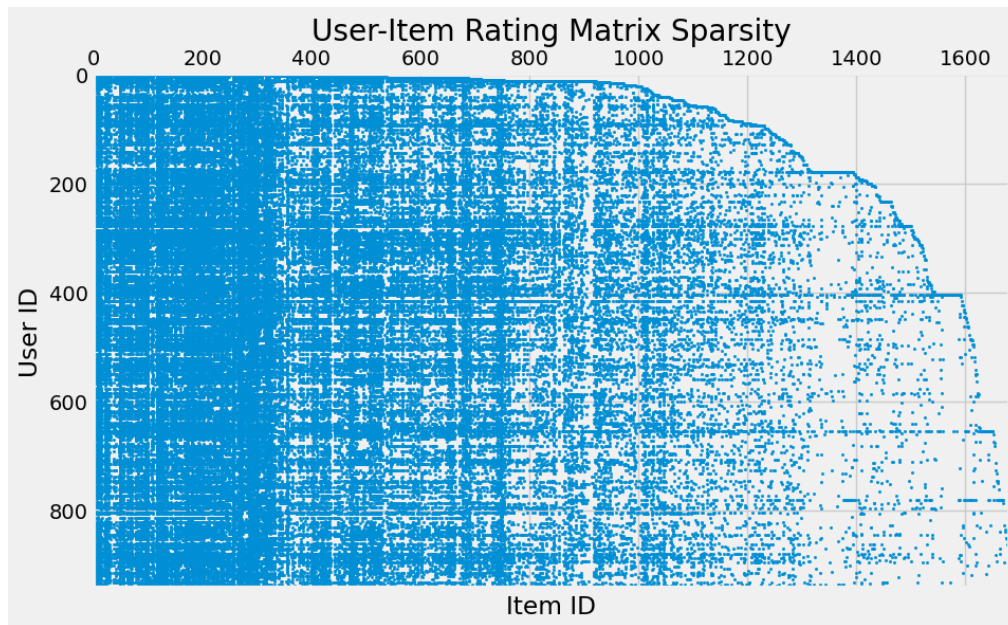


Figure 5: User-Item Matrix Sparsity Pattern

The matrix is relatively sparse, with a considerable part of the space unfilled. This indicates that collaborative filtering must be robust to missing data and highlights the importance of similarity-based or matrix factorization approaches that can operate in sparse domains.

We calculated the average rating given by each user and plotted a histogram to examine how user-level rating preferences are distributed.

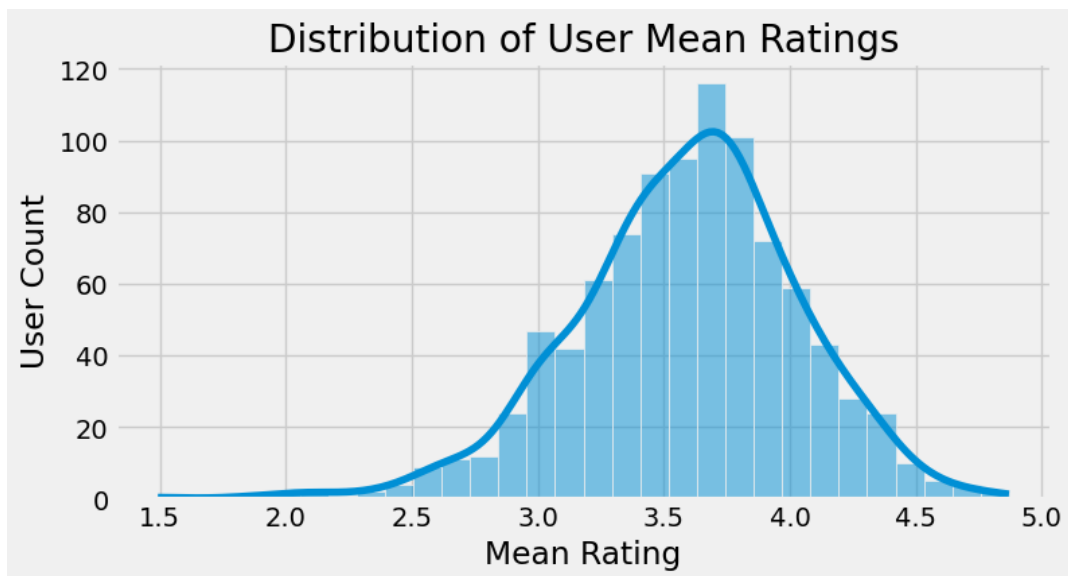


Figure 6: Distribution of User Mean Rating

The mean ratings are concentrated between 3.2 and 4.0, indicating that most users tend to rate movies positively. This variation suggests that users have different baselines or personal scales of rating, which justifies the need for mean-centering when computing similarities between users in collaborative filtering.

1.4 Result and Analysis

The 10 most similar users to user 1 are: [773, 868, 592, 880, 429, 276, 916, 222, 457, 8]. Among these, only 5 users provided ratings for item 508, with the following values:

Table 1: Ratings for Item 508 by User 1's Top Similar Users

User ID	Rating for Item 508
592	5.0
880	4.0
429	4.0
276	5.0
222	3.0

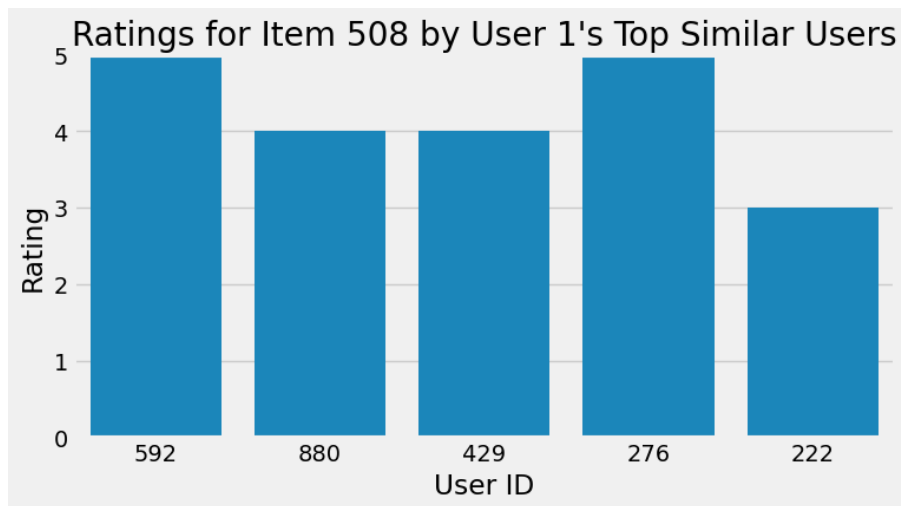


Figure 7: Top 10 Similar Users to User 1

The average of these ratings is:

$$\frac{5.0 + 4.0 + 4.0 + 5.0 + 3.0}{5} = 4.20$$

Thus, the predicted rating for user 1 on item 508 is **4.20**.

This prediction is derived purely from the preferences of similar users, assuming that users with similar rating behavior are likely to have similar tastes. The approach also handles sparsity by relying only on available neighbor ratings, which makes it suitable for collaborative filtering on sparse datasets like MovieLens.

2 Solution to Problem2

2.1 User Profile Similarity

To evaluate which user is more likely to enjoy a specific item (Movie 95), we implemented the following steps:

- First, we constructed a user-item utility matrix using the available rating data, where rows represent users and columns represent items.
- Then, we normalized the matrix by subtracting the mean rating of each user from their individual ratings. This centering process reduces user bias and allows for fairer comparisons.
- We extracted the centered rating profiles of two users of interest: User 200 and User 15.
- We also extracted the rating vector for Movie 95, representing how all users rated this item (after centering).
- To compare user profiles with the item vector, we aligned the vectors by selecting only the items with valid ratings in both vectors. This step avoids issues caused by missing values.
- After alignment, we computed the cosine similarity and cosine distance between each user's profile and the item vector. These metrics quantify how similar the users preferences are to the pattern of ratings associated with Movie 95.
- Finally, based on the similarity scores, we determined which user is more aligned with the item, and therefore more likely to receive a recommendation for Movie 95.

2.2 Results and Analysis

The computed similarity and distance values are summarized in Table 2.

Table 2: Cosine similarity and distance between user profiles and movie 95

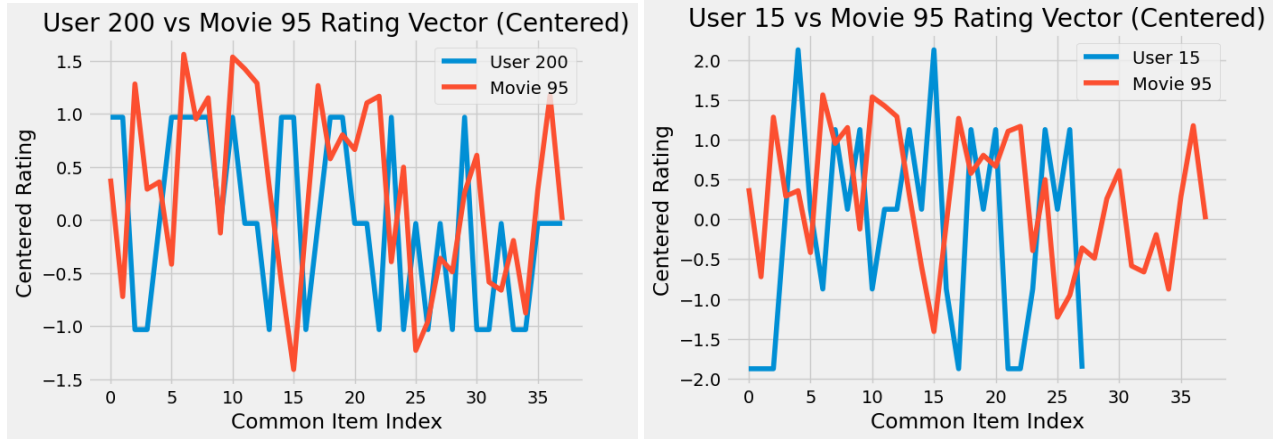
User	Cosine Similarity	Cosine Distance
User 200	0.0890	0.9110
User 15	0.1305	0.8695

From the results, user 15 exhibits a higher cosine similarity with movie 95 compared to user 200, indicating a closer alignment in rating preferences. Consequently, the recommender system would be more inclined to suggest movie 95 to user 15.

2.3 Visualization and Analysis of User-Item Relationships

To better understand the relationship between users and item 95, we visualize the centered rating vectors and their correlation through line plots and scatter plots.

The Figure 8 displays the centered ratings of User 200 and User 15 and item 95 across the set of commonly rated items. Each point on the x-axis corresponds to a shared item, while the y-axis represents the centered rating value (original rating minus user's average rating).



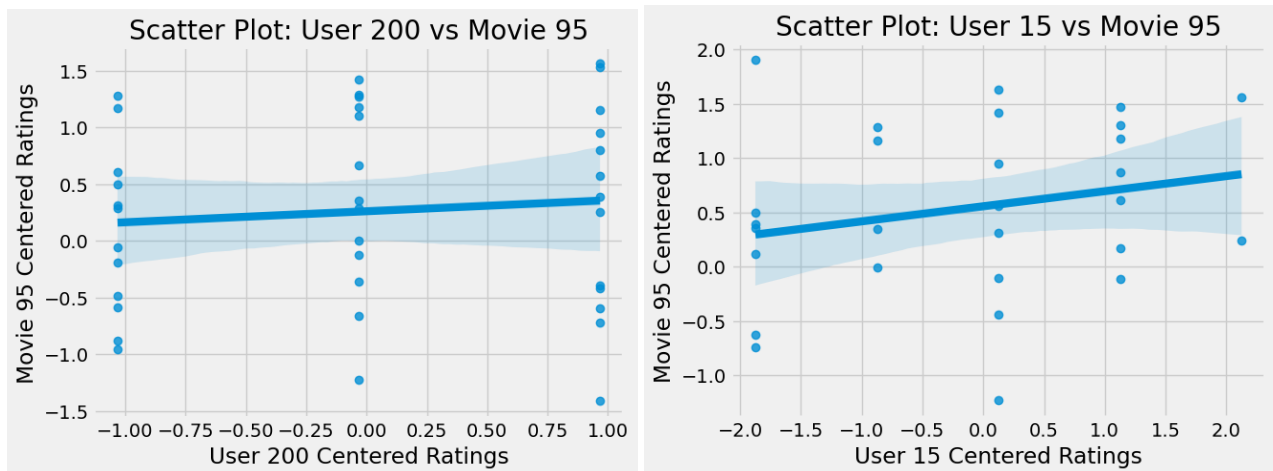
(a) Line Plot: User 200 vs Movie 95

(b) Line Plot: User 15 vs Movie 95

Figure 8: Line Plot

- Figure 8a shows that the rating patterns of User 200 and Movie 95 do not strongly align. Although there are points where their values move in the same direction, overall the fluctuations diverge. This observation is consistent with the low cosine similarity of 0.0890 between them.
- Figure 8b compares the centered ratings of User 15 and item 95. Compared to User 200, User 15's rating trend appears to be slightly more aligned with that of Movie 95. There are more overlapping peaks and troughs between the two curves. This supports the higher cosine similarity value of 0.1305, indicating a closer relationship.

Scatter plots in Figure 9 with a regression line is used to visualize the correlation between the centered ratings of User 200 and User 15 and item 95. Each dot represents a common item, plotted with User's rating on the x-axis and Movie 95's rating on the y-axis.



(a) Scatter Plot: User 200 vs Movie 95

(b) Scatter Plot: User 15 vs Movie 95

Figure 9: Scatter Plot

- Figure 9a reveals a weak linear relationship, with the data points being loosely scattered around the regression line. This further confirms the low similarity between User 200 and item 95.
- Figure 9b shows a slightly more defined positive trend compared to User 200, with points more concentrated around the regression line. This visual evidence supports the conclusion that User 15's preferences are more aligned with Movie 95 than User 200s, making User 15 a better recommendation target for this item.

2.4 Conclusion

By centering user ratings and computing cosine similarity, we can quantify how well a particular item matches a user's preferences. This approach effectively identifies user 15 as the more suitable candidate for recommending movie 95 based on their historical rating behavior.